commands—this is more than the **four** new command we had to learn in Section 1.11 but still an easily manageable task.

So we set ourselves in this chapter a modest goal, using `beamer` "out of the box". It is amazing how much you can achieve with a small investment of your time.

Chances are `beamer` is already installed for you. If not, consult Section D.1 on how to get it.

12.1 *Quick and dirty* `beamer`

We convert the article `firstarticle.tex` (in the `samples` folder) to a `beamer` presentation. We will remove some commands that are appropriate for an article but not for a presentation and add some commands that are specific to presentations. This will not produce a very good presentation. Nevertheless, the conversion is a really quick introduction to some basic `beamer` concepts.

12.1.1 *First changes*

Open `firstarticle.tex`, save it as `quickbeamer.tex` in the `work` folder. The converted `tex` version and the presentation `quickbeamer.pdf` are both in the `samples` folder.

Make the following changes in the preamble and top matter:

1. Change the first line to
   `% Introductory beamer presentation: quickbeamer.tex`
2. Change the documentclass to `beamer`.
3. Delete the six `\usepackage` and `\newtheorem` lines—`beamer` loads the necessary packages and defines these declarations.
4. Change the `\address` to `\institute`—this is the `beamer` command for address.
5. Delete the `abstract` environment—this is not needed for the presentation.

Here is the new version of the preamble and top matter.

```
%Introductory beamer presentation: quickbeamer.tex
\documentclass{beamer}
\begin{document}
\title{A construction of complete-simple\\
       distributive lattices}
\author{George~A. Menuhin}
\institute{Computer Science Department\\
         University of Winnebago\\
         Winnebago, MN 53714}
```

```
\date{March 15, 2015}

\maketitle
```

12.1.2 Changes in the body

1. Delete the `notation` and `proof` environments, but not the contents, that is, delete the four lines

   ```
   \begin{notation}
   \end{notation}
   \begin{proof}
   \end{proof}
   ```

   Both environments could theoretically stay, but the `notation` environment is not needed since in the next step we put `Notation` in the frame title. The `proof` environment (see Section 3.3.2) is not suitable for presentations because an environment can only be used within a frame, and proofs are typically longer.

2. Cut the presentation into *frames* (pages, transparencies, foils) with the `frame` environments. After each `\begin{frame}` we put a `\frametitle` command. The argument of the command is the "title" for the frame, displayed prominently at the top of the display.

   It would be tedious to give you precise instructions on how to do this, instead refer to the `quickbeamer.tex` document (in the `samples` folder) for all the `frame` environments and `\frametitle` commands we added.

3. Cut out the `figure` environment, except for the line

   ```
   \centering\includegraphics{products}
   ```

   which should be moved to follow the

   ```
   \frametitle{Illustrating the construction}
   ```

   line and accordingly delete `(see Figure~\ref{Fi:products})`.

Now copy over the illustration `products` from the `samples` folder to the `work` folder and typeset. That's it, enjoy your first presentation.

12.1.3 Making things prettier

Now you make some small changes to `quickbeamer.tex` to utilize `beamer`'s power for wonderful effects. Changes 1 and 3 are quite dramatic.

Save `quickbeamer.tex` with the name `quickbeamer1.tex` in the `work` folder. The edited version,`quickbeamer1.tex`, is in the `samples` folder along with the presentation `quickbeamer1.pdf`.

1. Add `\usetheme{Berkeley}` after the documentclass line.

2. Change `\maketitle` to

```
\begin{frame}
  \titlepage
\end{frame}
```

   Make sure that the last (sub)section is followed by a frame, otherwise it will be missing from the table of contents.

3. Add this frame after the titlepage frame:

```
\begin{frame}
\frametitle{Outline}
\tableofcontents[pausesections]
\end{frame}
```

   This creates a table of contents frame, with the section titles appearing one at a time.

4. Replace all instances of `{equation}` by `{equation*}`. In a presentation a reference to another frame is not recommended so equations should not be numbered. You might as well delete all the `\label` commands since these are not needed either.

5. In the second to last frame there are two references to equation numbers. Replace the text `the congruence \eqref{E:cong2}` with `the congruence`, and also replace `\eqref{E:comp}` with `the penultimate equation`, or similar.

6. Change the bibliographic reference to
   `See also Ernest T. Moynahan, 1957.`

Turning `quickbeamer1.tex` into a PDF file will get you a much prettier presentation. The first four pages of the new presentation are displayed in Figures 12.1 and 12.2—unfortunately, without the pretty colors.

12.1.4 Adjusting the navigation

Looking at Figures 12.1 and 12.2, we see that the `Berkeley` theme turns the sidebar into a navigation device and the section titles produce the table of contents (the Outline frame). Remember to typeset a few times! But a number of problems come to light.

1. The title of the presentation is too long for the sidebar, so is the title of the second section.

2. There is no need to repeat the author's name in the sidebar.

3. It was natural for the article `firstarticle.tex` to have only two sections. But sections play a different role in a `beamer` presentation, they are navigation tools.

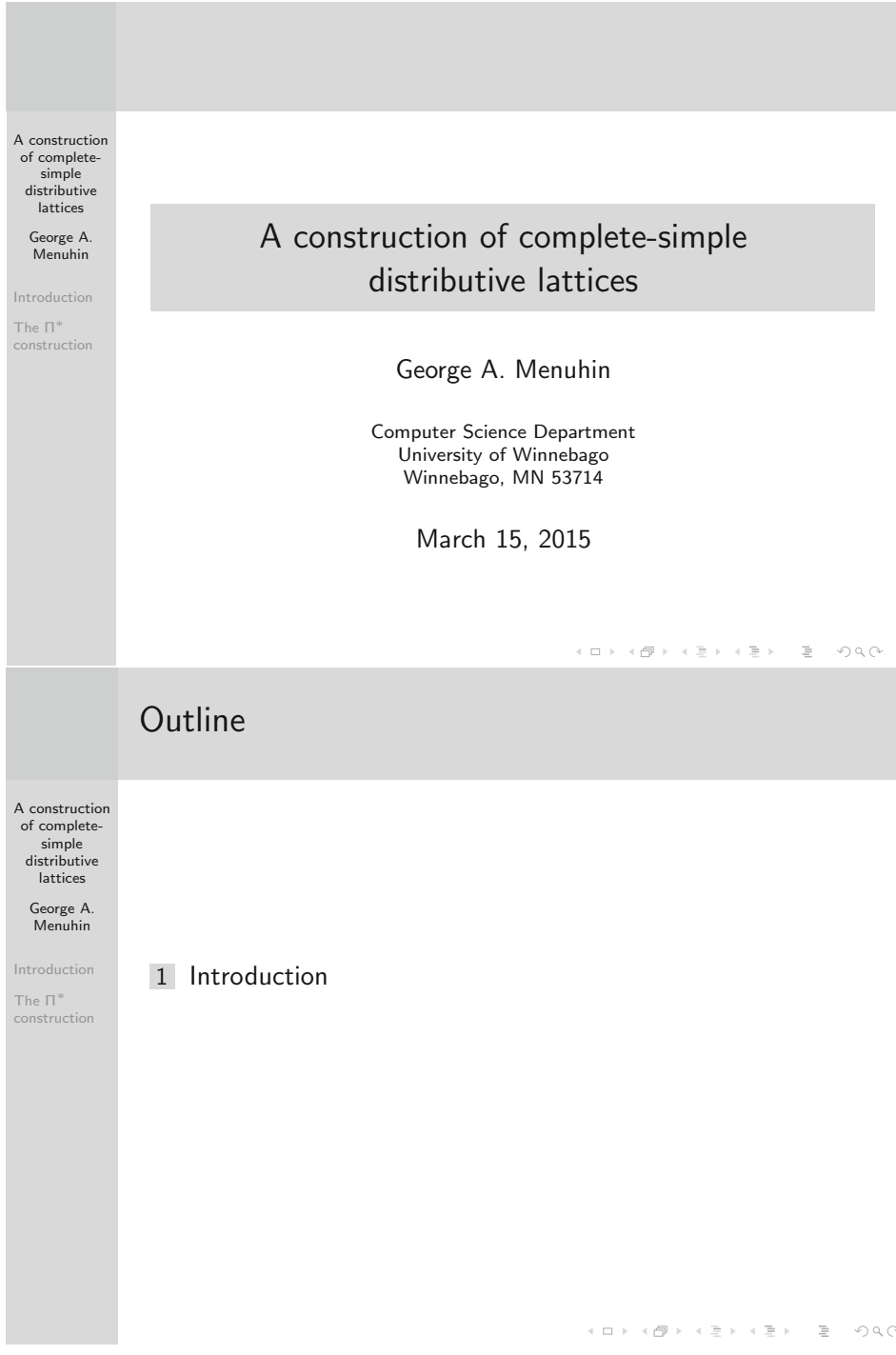


Figure 12.1: `quickbeamer1` presentation, pages 1 and 2.

Outline

1 Introduction

2 The Π* construction

Introduction

In this note, we prove the following result:

Theorem

There exists an infinite complete distributive lattice K with only the two trivial complete congruence relations.

Figure 12.2: `quickbeamer1` presentation, pages 3 and 4.

The sidebar lists all the sections. It also highlights the section we are in. Moreover, by clicking on the name of a section, the presentation jumps there.

To correct these deficiencies, save the file `quickbeamer1.tex` as `quickbeamer2.tex` in the `work` folder. The edited version is in the `samples` folder along with the PDF file.

1. Change the `\title` command to

   ```
   \title[Complete-simple distributive lattices]%
   {A construction of complete-simple\\
   distributive lattices}
   ```

   and the second `\section` command to

   ```
   \section[Construction]{The $\Pi^{*}$ construction}
   ```

   The bracketed parts are the short versions used in the sidebar.

2. Change the `\author` command to

   ```
   \author[]{George~A. Menuhin}
   ```

   The short version of the `author` command is blank, so the author's name will not be displayed in the sidebar.

3. Add the command

   ```
   \section[Second result]{The second result}
   ```

   before the frame of the same title and

   ```
   \section{Proof}
   ```

   before the proof. We even add

   ```
   \section{References}
   ```

   before the frame of the same name.

Figure 12.3 shows page 7 of the `quickbeamer2` presentation—this corresponds to page 4 of the `quickbeamer1` presentation; the Outline accounts for the difference. Note how all the deficiencies listed above have been corrected. Compare page 7 of this presentation with the `Berkeley` theme in Figure 12.3 and with the `Warsaw` theme in Figure 12.4. Themes are discussed in Section 12.5.

12.2 *Baby* `beamers`

In the previous discussions you may have noticed two interesting features. First, the Outline frame (table of contents) created *two* pages in the `quickbeamer1` presentation and *five* pages in the `quickbeamer2` presentation. We discuss this in some detail now with the `babybeamer` presentations. You can find all the `babybeamer` presentations as `tex` and PDF files in the `samples` folder. Second, the sidebar shows some links. More about this in Section 12.2.7.

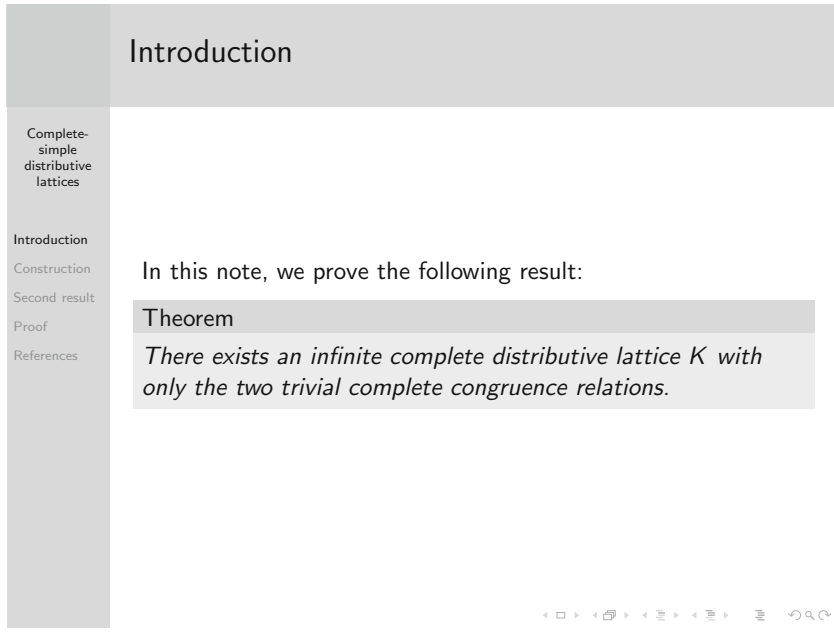


Figure 12.3: `quickbeamer2` presentation, page 7 with `Berkeley` theme.

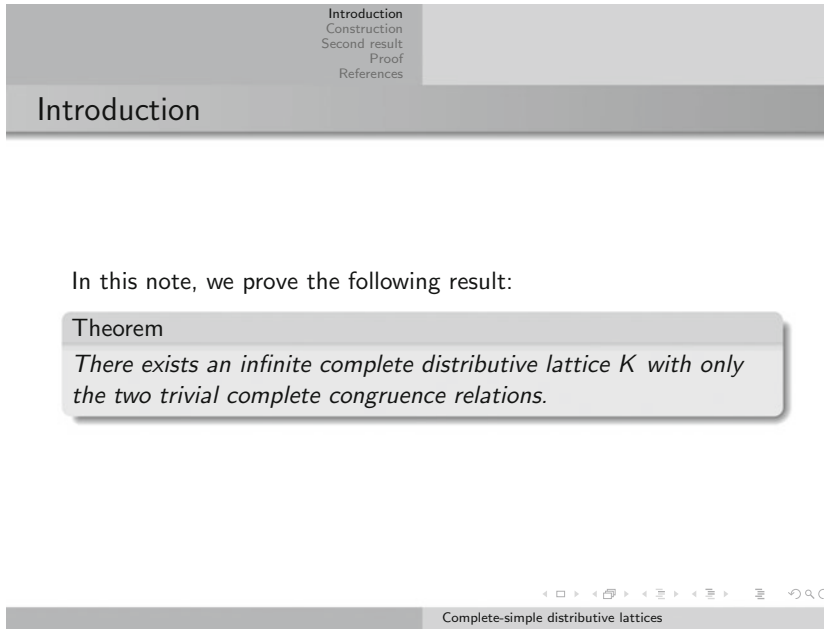


Figure 12.4: `quickbeamer2` presentation, page 7 with `Warsaw` theme.

12.2.1 Overlays

The outline frame of the quickbeamer2 presentation created five pages in the PDF file. Observe how each page, from the second on, completely overlaps the previous one, making it appear that the previous one stayed put and an additional line is displayed "on top of it". In beamer terminology these pages are *overlays* or *slides*. The five overlays will be referenced as overlay 1, ..., overlay 5. A single frame may create one or many overlays. The subsequent sections discuss many more variants.

beamer has many commands creating overlays. We start with some examples of \pause, then \only, and \onslide.

We introduce overlays with some presentations. The first, babybeamer1, introduces the \pause command to create overlays.

```
%babybeamer1 presentation
\documentclass{beamer}
\begin{document}

\begin{frame}
\frametitle{Some background}

We start our discussion with some concepts.
\pause

The first concept we introduce originates with Erd\H os.
\end{frame}
\end{document}
```

produces the presentation of Figure 12.5.

Rule ■ The \pause command

1. A frame may have many \pause commands.

2. The \pause command cannot be given in an AMS multiline math environment.

You move past a \pause command the same way as you get to the next frame, by pressing the space bar or the forward arrow key.

Using the \pause commands you can create many overlays, each containing a little more material on the overlays. If this is all you need, skip to Section 12.2.7, you do not need the more detailed discussion of overlays in the next few pages.

We could have coded the same presentation with the \only command:

```
%babybeamer2 presentation
\documentclass{beamer}
```

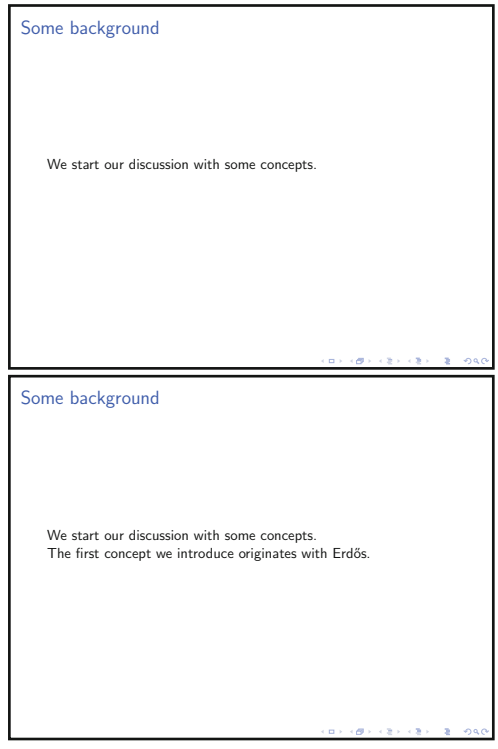


Figure 12.5: `babybeamer1` presentation.

```
\begin{document}

\begin{frame}
\frametitle{Some background}

\only<1,2>{We start our discussion with some concepts.}

\only<2>{The first concept we introduce originates
         with Erd\H os.}
\end{frame}
\end{document}
```

This presentation is slightly different from `babybeamer1`. Overlay 1 ignores the second `\only` command and displays the line as appropriate to display one line. Overlay 2 displays the two lines as appropriate to display two lines. As a result, the first line moves slightly up when passing from overlay 1 to overlay 2. The argument of the `\only` command is typeset only on the overlays specified. On the other overlays, it is ignored.

If instead of the \only command you use the \onslide command (on slide, get it?), as in

```
\onslide<1,2>{We start our discussion with some concepts.}

\onslide<2>{The first concept we introduce originates
         with Erd\H os.}
```

then the first line of overlay 2 completely overlaps the first line of overlay 1, so the first line seems to stay put. The argument of the \onslide command is typeset on the overlays specified and on the other overlays it is typeset but invisible. This is the behavior you would want most often, but you may find that sometimes you prefer \only.

12.2.2 Understanding overlays

We introduced overlays in Section 12.2.1—probably the most important new concept for presentations. LaTeX typesets the content of a frame and the typeset material

- appears on all overlays for the parts of the source (maybe all) not modified by any command with an overlay specification;
- appears only on the overlays specified and is ignored on the other overlays for the arguments of the \only commands;
- appears on the overlays specified and is typeset but made invisible on the other overlays for the arguments of the \onslide commands.

More on overlay specifications at the end of this section. Here are some illustrations.

Example 1

```
This is a very \only<1>{very, very} important concept.
\only<1,2>{To start the definition \dots}
```

will typeset overlay 1 as

This is a very very, very important concept. To start the definition ...

and will typeset overlay 2 as

This is a very important concept. To start the definition ...

Example 2

```
What is $2+2$? It is \onslide<2>{$4$}.
\only<1>{Can you figure it out?}

\onslide<2>{I hope you all got it right.}
```

will typeset overlay 1 as

What is $2+2$? It is . Can you figure it out?

and will typeset overlay 2 as

What is $2+2$? It is 4.
I hope you all got it right.

Note that there is room in overlay 1 for the number 4.

Example 3

```
What is $2+2$?

\onslide<2>{It is {$4$}.}

Can you figure it out?
```

will typeset overlay 1 as

What is $2+2$?

Can you figure it out?

and will typeset overlay 2 as

What is $2+2$?
It is 4.
Can you figure it out?

Note that there is room in overlay 1 for the "missing" second line.

Overlay specifications

The angle brackets contain the *overlay specification*. Here are some more examples:

`<1-2,4->` means all overlays from 1 to 2, and all overlays from 4 onwards.
`<-3>` means all overlays up to 3.
`<2,4,6>` means overlays 2, 4, and 6.

We have two overlay specifications in the presentation `babybeamer2`: `<1,2>` and `<2>`. Maybe, `<1->` and `<2->` would be better, so that if you add a third overlay you do not have to change these.

The command `\pause` can only take the simplest overlay specification, a number. `\pause<3>` takes effect from overlay 3 on.

Note that overlay specifications are attached to commands but the overlays created are overlays of the frame in which the commands appear.

12.2.3 More on the `\only` and `\onslide` commands

The `\only` and `\onslide` commands can accomplish everything the `\pause` command can and a lot more.

The basic syntax

The syntax of `\only` is

`\only<`*overlay spec*`>{`*source*`}`

where *overlay spec* is the overlay specification and *source* is the code typeset by LaTeX.

A (partial) syntax of `\onslide` is

`\onslide<`*overlay spec*`>{`*source*`}`

With the same syntax you can give overlay specifications to many commands, including `\textbf`, `\textit`, `\alert`—`beamer`'s alternative to the `\emph` command—and then the command is in effect only on the overlays specified.

```
%babybeamer3 presentation
\documentclass{beamer}
\begin{document}

\begin{frame}
\frametitle{Some background}

\textbf<1>{We start our discussion with some concepts.}

\textbf<2>{The first concept we introduce originates
           with Erd\H os.}
```

```
\end{frame}
\end{document}
```

So the `babybeamer3` presentation (see Figure 12.6) has two overlays, each with two lines of text. On overlay 1 the first line is bold, on overlay 2 the second line is bold.

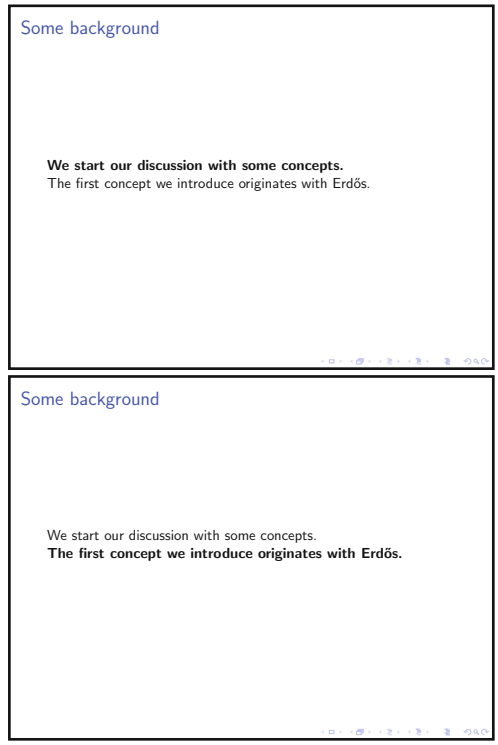


Figure 12.6: `babybeamer3` presentation.

A different syntax

The command `\only` has an alternate syntax:

`\only{`*source*`}<`*overlay spec*`>`

So

```
\only<1>{Can you figure it out?}
```

and

```
\only{Can you figure it out?}<1>
```

accomplish the same.

With this syntax, you can define your own commands that allow overlay specifications. For instance, using the command `\color{blue}` defined in Section 12.2.9, you can define the command

```
\newcommand{\myblue}{\only{\color{blue}}}
```

Then

```
\myblue<2>{Some more text}
```

will color the text blue on overlay 2 only.

12.2.4 Lists as overlays

Lists may be presented one item at a time, for example the `babybeamer4` presentation in Figure 12.7 (in the `samples` folder) shows the four overlays of a list. R. Padmanabhan appears on the first, R. Padmanabhan and Brian Davey appear on the second, and so on. This is accomplished simply by adding the overlay specification `<1->` to the item for R. Padmanabhan, the overlay specification `<2->` to the item for Brian Davey, and so on.

```
%babybeamer4 presentation
\documentclass{beamer}
\begin{document}

\begin{frame}
\frametitle{Overlaying lists}

We introduce our guests:
\begin{itemize}
\item<1-> R. Padmanabhan
\item<2-> Brian Davey
\item<3-> Harry Lakser
\item<4-> Dick Koch
\end{itemize}
\end{frame}
\end{document}
```

Such an overlay structure is used so often that `beamer` has a shorthand for it, `[<+->]`. Here it is in `babybeamer5`.

```
%babybeamer5 presentation
\documentclass{beamer}
\begin{document}
```

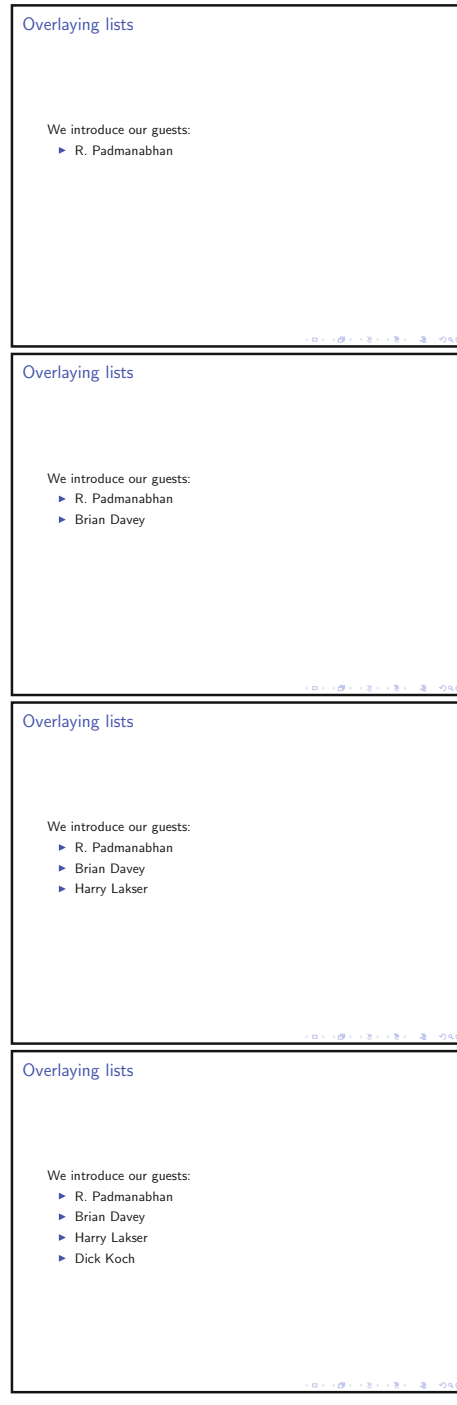


Figure 12.7: `babybeamer4` presentation.